

Programme	B. Sc. Computer Science				
Course Title	Parallel Computing				
Type of Course	Major				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Knowledge in Fundamentals of COA and Operating System				
Course Summary	The syllabus is prepared with the view of preparing the BSc Computer Science Graduates to understand basic and advanced concepts of parallel computing. It covers Principles of Parallel Algorithm Design, Communication operations, Programming Using the Message Passing Paradigm, Programming Shared Address Space Platforms, Thread Basics,				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Summarize the key parallel computational models	U	C	Instructor-created exams / Quiz
CO2	Appreciate and apply parallel and distributed algorithms in problem Solving	U	C	Instructor-created exams / Quiz
CO3	Appreciate the communication models for parallel algorithm development	U	P	Instructor-created exams / Quiz
CO4	Develop parallel algorithms using message passing paradigm	U	C	Instructor-created exams / Case studies
CO5	Formulate parallel algorithms for shared memory architectures	U	P	Instructor-created exams / Quiz Case studies
CO6	Understand thread management	U	P	Instructor-created exams / Quiz /Case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
--------	------	---------	-----	-------

I	Principles of Parallel Algorithm Design		12	15
	1	Parallel Processing platforms. Preliminaries, Decomposition Techniques,	2	
	2	Characteristics of Tasks and Interactions	2	
	3	Mapping Techniques for Load Balancing	2	
	4	Methods for Containing Interaction Overheads	3	
	5	Parallel Algorithm Models.	3	
II	Communication Operations		12	15
	6	Basic Communication Operations - One-to-All Broadcast and All-to-One Reduction	2	
	7	All-to-All Broadcast and Reduction	1	
	8	All-Reduce and Prefix-Sum Operations	3	
	9	Scatter and Gather	3	
	10	All-to-All Personalized Communication, Circular Shift	1	
	11	Improving the Speed of Some Communication Operation	2	
III	Programming Using the Message Passing Paradigm		12	20
	12	Principles of Message-Passing Programming, The Building Blocks: Send Operations	2	
	13	Receive Operations	2	
	14	MPI: The Message Passing Interface	2	
	15	Overlapping Communication with Computation	2	
	16	Collective Communication and Computation Operations	2	
	17	Groups and Communicators	2	
IV	Programming Shared Address Space Platforms Thread Basics		12	20
	18	Thread Basics, Why Threads? The POSIX Thread Application Programme Interface, Synchronization Primitives in POSIX, Controlling Thread and Synchronization Attributes	2	
	19	Thread Cancellation, Composite Synchronization Constructs	2	
	20	OpenMP: a Standard for Directive Based Parallel Programming, Specifying Concurrent Tasks in OpenMP	3	
	21	Synchronization Constructs in OpenMP	2	
	22	OpenMP Applications: Parallel algorithm development for Matrix multiplication	3	
V	Open Ended Module		12	
	CASE STUDY: PARALLEL COMPUTING Heterogeneous Parallel Computing Data parallel computing Device Global Memory and Data Transfer Kernel Functions and Threading			

Reference Books:

1. Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar, Introduction to Parallel Computing, 2nd Ed, Addison-Wesley, 2003

2. David B. Kirk, Wen-mei W. Hwu, Programming Massively Parallel Processors: A Hands-on Approach, 3rd Ed., Morgan Kaufman, 2016. References
3. Steven Brawer, Introduction to Parallel Computing, Academic Press, (1989)
4. Barbara Chapman, Gabriele Jost, Ruud van der Pas, Using OpenMP: Portable Shared Memory Parallel Programming, MIT Press, 2008.
5. William Gropp, Ewing Lusk, Anthony Skjellum Using MPI: Portable Parallel Programming with the Message-Passing Interface, 3rd Ed, MIT Press, 2014.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO1	-	2	-	-	-	-						
CO2	-	2	1	-	-	-						
CO3	-	2	1	-	1	1						
CO4	-	2	1	1	1	2						
CO5	-	3	1	1	-	2						
CO6	-	2	-	-	-	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓

CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	